

TEAM PRXFR · IQOO CITY BATTLES 2026

WorldJam

Record the real sound of anything around you. On-device AI turns it into music.

Android · React Native

Gemma 4 on-device

Native C++ audio

Music apps hand you someone else's sounds

The same sample packs

Every beginner opens the same library and makes something that sounds like everyone else's.

A studio you must learn

A DAW is a wall of controls. The idea dies before the first bar is finished.

AI that sends audio away

Generative tools upload your recordings and need a connection to answer.

Meanwhile every room already has instruments in it: a ceramic cup, a wooden table, a set of keys.

THE IDEA

Anything can be a studio

01 · Capture

Hold and strike an object. The phone records its real sound and measures how bright, how long and how pitched it is.

02 · Direct

Pick a vibe, or type what you want — an artist, a feeling, an instrument. Sing, and the melody becomes the song.

03 · Compose

Gemma, running on the phone, arranges your sounds into a structured track with verses, builds and fills.

RUNNING ON DEVICE

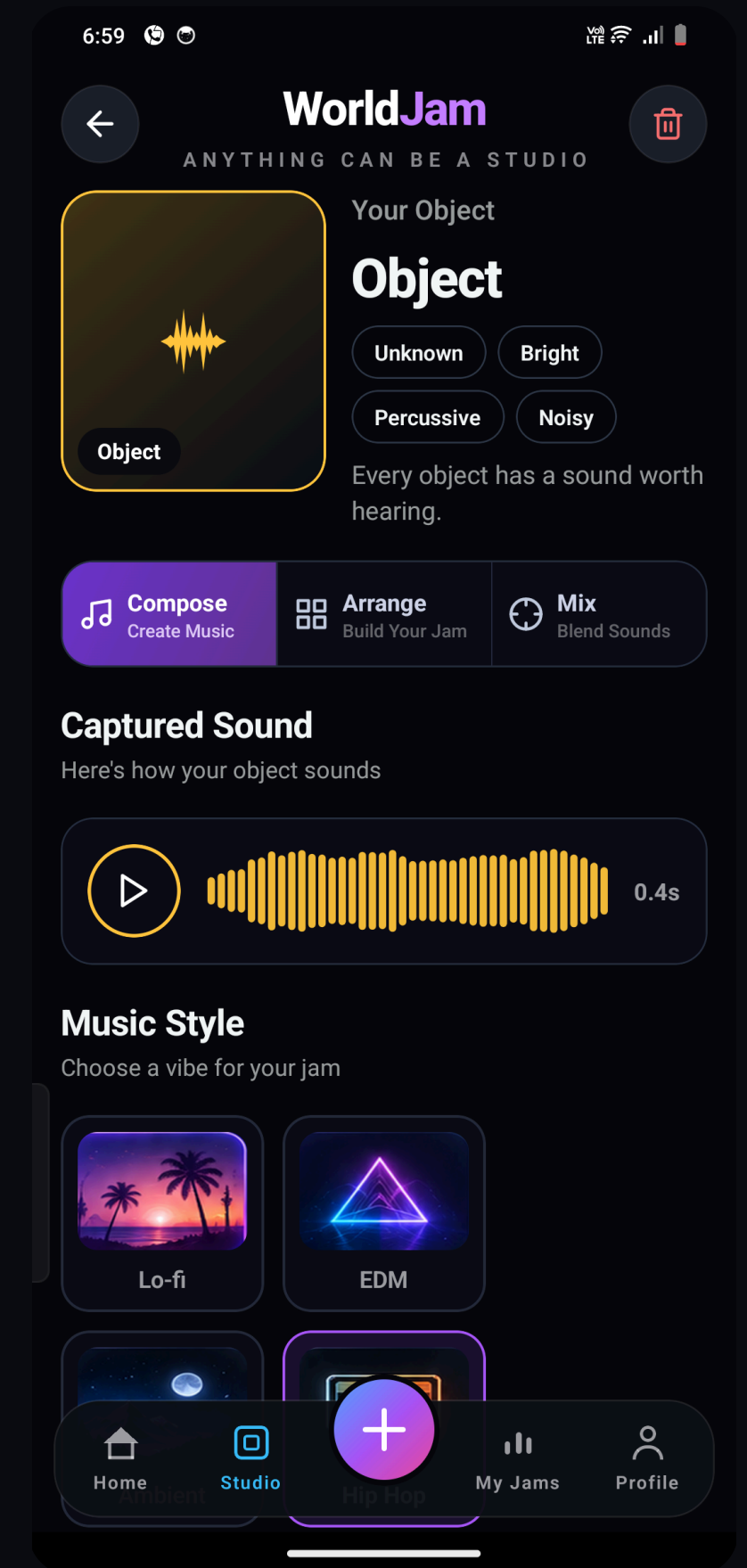
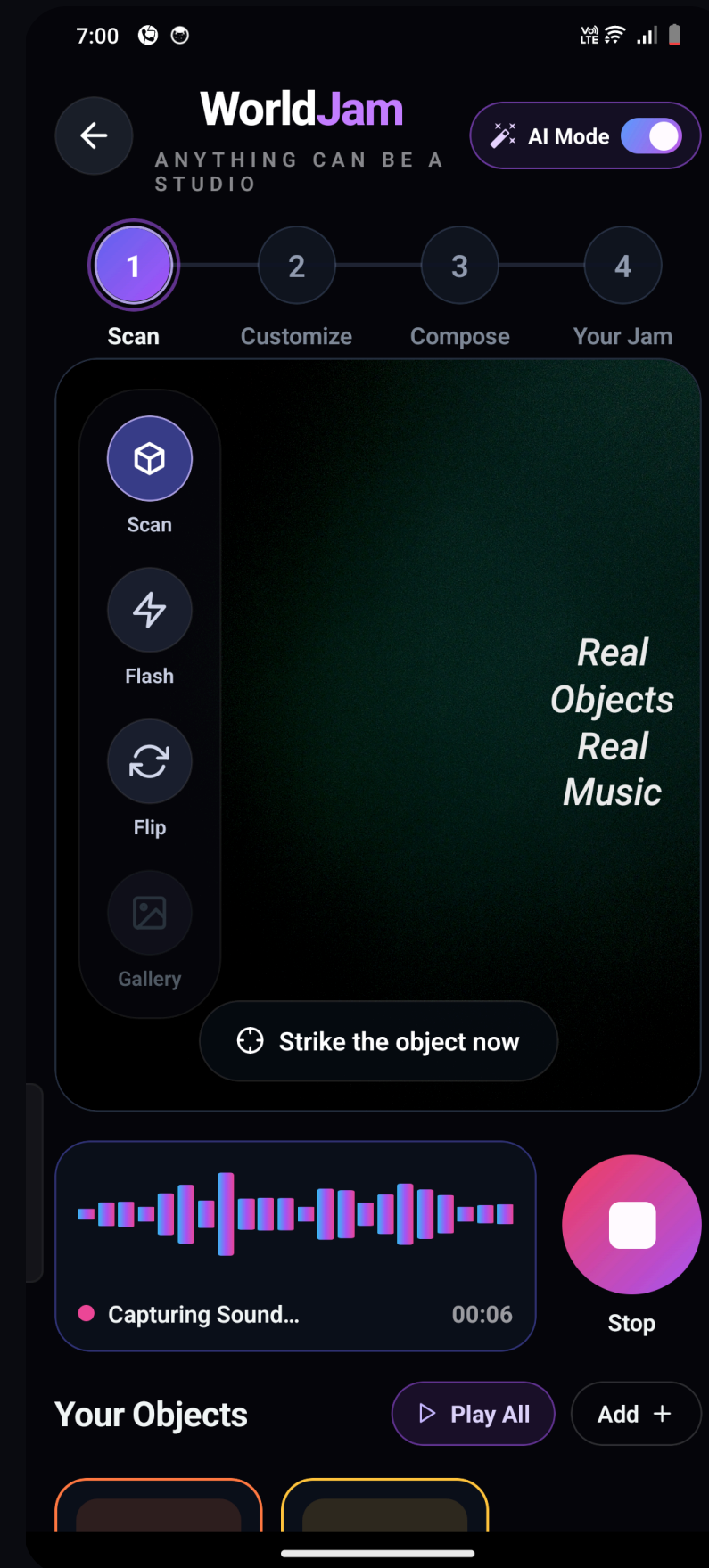
Four steps, one screen each

Scan — viewfinder, hold to record, your objects below

Customize — vibe, tempo, and a box that takes plain English

Compose — the arrangement, laid out on a timeline

Your Jam — transport, layer mixer, export



Three layers, each doing one job

Native audio

C++ and Oboe, 651 lines. A lock-free ring buffer carries taps to a callback that never allocates, locks or logs.

Signal analysis

FFT, YIN pitch tracking, onset detection and key estimation decide what each recording should be in the arrangement.

On-device model

Gemma 4 E2B via llama.cpp emits a strict JSON plan. A rule-based arranger answers instantly if it is slow.

MEASURED ON DEVICE

24.8 ms from tap to sound

Below the ~50 ms threshold where a tap stops feeling like an instrument and starts feeling like a button. This is the number the whole product rests on, so it is measured in the app rather than claimed.

Exclusive-mode Oboe stream

Look-ahead scheduler

Zero-allocation callback

The model runs on the phone. All of it.

Gemma 4 E2B, 3.35 GB

Loaded through llama.cpp in about 10 seconds. Prompted with the objects' measured features and, when you sing, a description of your melody.

No network path

Recordings never leave the device. There is no upload step to disable, because one was never built.

Degrades, never fails

Model slow, missing or wrong format? The rule-based arranger answers in microseconds and the app says which one you heard.

The runtime picks the chat template from the model file, so swapping in a different GGUF is a file copy, not a code change.

WHAT MAKES IT SOUND LIKE MUSIC

A loop repeats. A song goes somewhere.

Technique	Why it matters
Sections with per-role density	A verse thins parts to their strong beats instead of muting them, so it still sounds like the same song as the chorus.
Velocity follows the bar	Downbeat hardest, offbeats lightest. Flat velocity is the clearest sign something was sequenced, not played.
Fills on every turnaround	Deliberately quieter than the downbeat they lead into, so the arrival is not swallowed.
Sung melody drives the song	Hum a line: it is tuned to the scale, timed to its own tempo, and the arrangement is built around it.
No synthetic drums	Your objects are the percussion. Layering a synth kit over them is what makes an app sound like a backing track.

Playable without looking at the screen

Spoken guidance

The app describes what it hears and confirms each capture aloud, so a blind user can build a track by touch and sound alone.

Spatial audio

Objects are panned to where they sit in the room, so the mix is also a map of what is around you.

Labelled controls

Every control carries a screen-reader label and role, including the knobs and faders, which are adjustable by gesture.

Accessibility was a design input, not a pass at the end. An instrument you must watch is not an instrument.

Built and running, not a prototype

389

tests passing

24.8

ms tap to sound

9

screens shipped

0

bundled sounds

Capture, playback, arrangement, sung-melody analysis, lyrics, export, session saving and the full four-step flow all work on a real handset today. The last number is the point: every sound in a WorldJam track is one the user recorded.

Against the rubric

Criterion	Weight	What WorldJam brings
End product quality	30%	Runs on device now. 389 tests. Every screen reachable, every feature wired to real audio.
Novelty and impact	20%	No sample packs. The instrument is whatever is on your desk, and it works without sight or a network.
Creative phone use	15%	Microphone, camera, haptics and an on-device LLM are all load-bearing, not decoration.
Technical depth	15%	Custom C++ audio engine, lock-free scheduling, FFT and pitch analysis, a validated JSON contract with a fallback.
Office Kit usage	10%	Built and debugged phone-to-laptop throughout, with live device logs driving the fixes.
Demo and presentation	10%	The demo is the product: tap a cup on stage, then play back the song it became.

WHAT HAPPENS ON STAGE

Hand us any object. We will make a song out of it.

Three minutes: record two objects from the judging table, hum eight bars, and play back the track the phone composed — with no connection and nothing pre-loaded.

Team PRXFR

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